

# **MQAT**

## **Exercises in R**

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# 1. Quantitative Methods of Analysis in Transportation course

Materials, exercises, data and tutorials for the [Quantitative Methods of Analysis in Transportation](#) course of the [MSc in Transportation Systems](#) at Instituto Superior Técnico - University of Lisbon.



**TÉCNICO**  
UNIVERSIDADE  
DE LISBOA

**MESTRADO**  
*EM SISTEMAS DE  
TRANSPORTES*

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🔥 This is not the official book for the course.

Please refer to Fénix: [Quantitative Methods of Analysis in Transportation](#)



## 2. Introduction

This book aims to provide tools to deal with transportation datasets using [R programming](#), an open-source and widely used tool for data analytics in urban mobility.

It includes course sporting materials, exercises, and examples to explore with **R**, with a focus on modelling and spatial data analysis using GIS techniques.

### Why R and GIS

Most academic programs focus on teaching modelling and deep analysis of data. However, there is a need to learn how to explore and prepare a dataset for modelling. The use of **programming and GIS** techniques have enormous advantages, including their flexibility; reproducibility; and transparency and understanding the step-by-step process.

The use of GIS techniques in transportation is of enormous relevance when doing accessibility analysis or reeling with georeferenced transportation data, such as bike sharing route trips' datasets, origin-destination flows datasets, home/work locations, GTFS public transit data, and so on. There is a need to learn how to locate these open datasets, how to explore them and how to integrate them into transportation and urban analysis. Additionally, the use of open source software and datasets allows researchers to perform methods that are reproducible and transparent.

### TLDR

- Open-source tools widely used in data analytics and spatial analysis
- Flexibility and reproducibility in data manipulation and visualization
- Critical for urban mobility and transportation research, with spatial relevance
- Large transportation datasets are becoming increasingly common

### Recommended readings

- Engel (2023) [Introduction to R](#).
- Wickham et al. (2017) [R for Data Science](#).
- Lovelace et al. (2024) [Geocomputation with R](#).



## 3. Software

In this chapter we will guide you through the installation of R, Positron/RStudio IDE and the packages you will need for this course.

**R** and **Positron/RStudio**<sup>1</sup> are separate downloads.

### 3.1. R

You will need **R** installed on your computer. **R stats** (how it is also known) is a programming language and free software environment for statistical computing and graphics supported by the R Foundation for Statistical Computing.

The download links live at [The Comprehensive R Archive Network](#) (aka CRAN). The most recent version is 4.6.1, but you can use any `>= 4.1.x` if you already have it installed.

#### 3.1.1. Windows

[Download R for Windows](#) and run the executable file.

##### Important

You may also need to install [Rtools](#), which is a collection of tools necessary to build R packages in Windows.

#### 3.1.2. Mac

[Download R for MacOX](#). You will have to choose between the *arm64* or the *x86-64* version.

Download the `.pkg` file and install it as usual.

---

<sup>1</sup>We will use Positron. Although if you prefer RStudio or already use other IDE such as VScode, that's also fine.

### 3. Software

#### 3.1.3. Ubuntu

! These are instructions for Ubuntu. If you use other linux distribution, please follow the instructions on [The Comprehensive R Archive Network - CRAN](#).

You can look for R in the Ubuntu **Software Center** or install it via the terminal:

```
# sudo apt update && sudo apt upgrade -y
sudo apt install r-base
```

Or, if you prefer, you can install the latest version of R from CRAN:

```
# update indices
sudo apt update -qq
# install two helper packages we need
sudo apt install --no-install-recommends software-properties-common
dirmngr
# add the signing key (by Michael Rutter) for these repos
wget -qO- https://cloud.r-project.org/bin/linux/ubuntu/marutter_pubkey.asc
| sudo tee -a /etc/apt/trusted.gpg.d/cran_ubuntu_key.asc
# add the R 4.0 repo from CRAN
sudo add-apt-repository "deb https://cloud.r-project.org/bin/linux/ubuntu
$(lsb_release -cs)-cran40/"
```

#### ! Using an Ubuntu based distribution, but not pure Ubuntu?

If you are using an Ubuntu based Linux distribution, such as Linux Mint, the last command from the previous script will set the CRAN repository to an invalid address, causing the next step to throw an error when trying to fetch and install R.

To fix it, run the commands below.

```
# Remove invalid repository defined before
sudo add-apt-repository --remove "deb
https://cloud.r-project.org/bin/linux/ubuntu $(lsb_release
-cs)-cran40/"
# Set repository to fixed address, considering Ubuntu version
(upstream) instead of your Ubuntu flavour one (which is invalid
for CRAN)
sudo add-apt-repository "deb
https://cloud.r-project.org/bin/linux/ubuntu $(source
/etc/upstream-release/lsb-release && echo
$DISTRIB_CODENAME)-cran40/"
```

Then run:

```
sudo apt install r-base r-base-core r-recommended r-base-dev
```

### Optional

To keep up-to-date R version and packages, you can follow the instructions at [r2u](#).

After this installation, you don't need to open R base. Please proceed to install the IDE.

## 3.2. IDE

An IDE (Integrated Development Environment) is a software that provides tools to write, test and execute code in a single graphical interface. It makes the process of programming easier and more accessible to users less proficient in coding.

Below are tutorials to install two alternatives that support R. Both include a console, syntax-highlighting editor that supports direct code execution, as well as tools for plotting, history, debugging and workspace management. Positron has a more modern UI, but RStudio is more established due to its longevity. You can choose either.

### 3.2.1. Positron

Positron is available for free download from [Posit Positron](#).

### 3.2.2. Windows

[Download Positron](#) for Windows and run the executable file.

#### User or system install?

You can choose either. User install will keep Positron available only for your user, while System install will make it available to any user on your computer.

#### Is Positron failing to start R session?

If after successfully installing Positron you open it and face an error on the R console, installing Microsoft Visual C++ Redistributable can resolve missing dependencies that might be causing the problem.

To install it, open [this link](#) and download the **.exe** file in the **Link** column that suits your system architecture. Click on the file and run the installation procedure. Restart Positron. The issue should be gone.

### 3. Software

#### 3.2.3. MacOS

Download [Positron](#) for macOS file and install it as usual.

#### 3.2.4. Ubuntu

Download [Positron](#) .deb file and install it as usual.

#### 3.2.5. RStudio

RStudio is available for free download from [Posit RStudio](#).

#### 3.2.6. Windows

Download [RStudio IDE](#) .exe file and run the executable file.

#### 3.2.7. MacOS

Download [RStudio IDE](#) .dmg file and install it as usual.

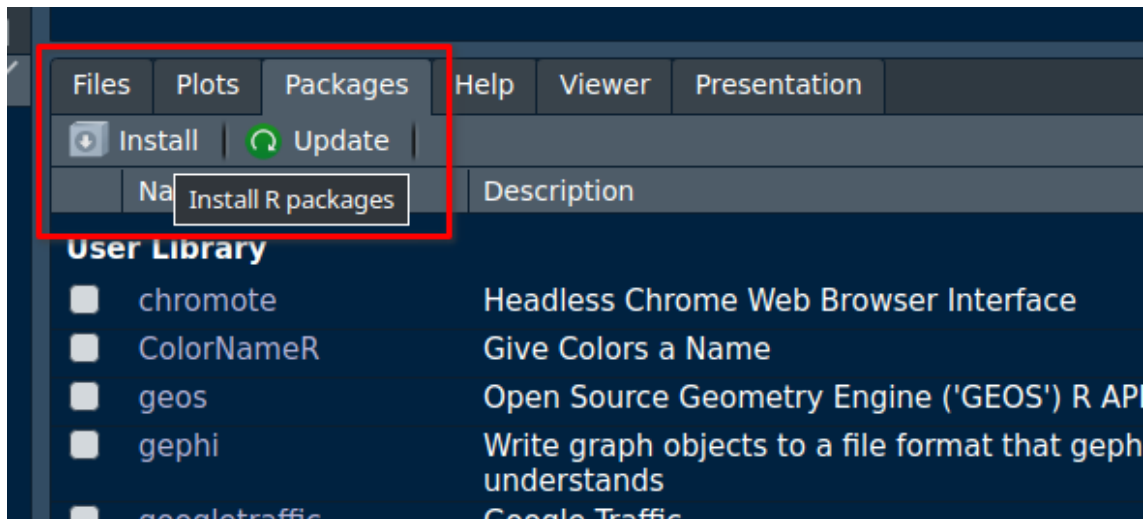
#### 3.2.8. Ubuntu

Download [RStudio IDE](#) .deb file and install it as usual.

### 3.3. R packages

You will need to install some packages to work with the data and scripts in this course.

You can install them in RStudio by searching for them in the **Packages** tab:



or by running the following code in the console:

```
install.packages("tidyverse")
install.packages("readxl")

install.packages("sf")
install.packages("mapview")
install.packages("rmarkdown")
install.packages("centr")
install.packages("od")
install.packages("openrouteservice")

install.packages(c("remotes", "devtools", "usethis")) # optional
install.packages("osmextract") # optional
install.packages("stplanr") # optional
```

#### ⚠️ Facing problems installing packages on Linux?

It is possible that the installation of the packages above fails due to missing dependencies. Unlike Windows or Mac, Linux sometimes tries to “build” packages from scratch, which causes errors if certain system tools are missing. To avoid this, follow the steps below to tell R to download ready-to-use packages instead of trying to build them.

1. On the **Terminal** (next to the console, at the bottom of your screen, there is a tab called **Terminal**), run the command below to get your OS **codename**:

```
# If you are using an Ubuntu original distro, get the value at
'codename'
lsb_release -a
# If you use other Ubuntu based distro (such as Linux Mint), get
the value returned by the command below
source /etc/upstream-release/lsb-release && echo $DISTRIB_CODENAME
```

2. Back on the **Console**, run the following R code, making sure to replace ‘resolute’ with the codename retrieved in the previous step.

```
# Use pre-compiled binary packages to avoid errors due to missing
dependencies
options(
  repos = c(CRAN =
    "https://packagemanager.posit.co/cran/__linux__/resolute/latest"),
  HTTPUserAgent = sprintf("R/%s R (%s)", getRversion(),
    paste(getRversion(), R.version["platform"],
    R.version["arch"], R.version["os"])))
)

# Disable bspm interception (use pre-compiled Linux binaries
  directly from Posit without needing apt or system privileges)
# install.packages(c("bspm"))
bspm::disable()
```

3. All set! Try to install the packages again. It should run smoothly now. :)

## 3.4. Cheat sheets

**i** This is the only cheating allowed at this course.

Cheath sheets are a concise set of notes used for quick reference. Posit, the open-source data

science software corporation responsible for RStudio/Positron IDEs, also makes available several cheat sheets that provide a comprehensive introduction to many of R softwares and packages.

You can start by taking a look at the ones for [Positron](#) or [RStudio](#).

Refer to [rstudio.github.io/cheatsheets](https://rstudio.github.io/cheatsheets) for the full cheat sheet repository.



**Part I.**

**Data and Models**



## 4. R basics

In this chapter we will introduce to the R basics and some exercises to get familiar to how R works.

### 4.1. Math operations

#### 4.1.1. Sum

```
1+1
```

```
[1] 2
```

#### 4.1.2. Subtraction

```
5-2
```

```
[1] 3
```

#### 4.1.3. Multiplication

```
2*2
```

```
[1] 4
```

#### 4.1.4. Division

```
8/2
```

```
[1] 4
```

## 4. R basics

### 4.1.5. Round the number

```
round(3.14)
```

```
[1] 3
```

```
round(3.14, 1) # The "1" indicates to round it up to 1 decimal digit.
```

```
[1] 3.1
```

You can use `help ?round` in the console to see the description of the function, and the default arguments.

## 4.2. Basics

### 4.2.1. Combinations

```
c(1, 2, 3)
```

```
[1] 1 2 3
```

```
c(1:3) # The ":" indicates a range between the first and second numbers.
```

```
[1] 1 2 3
```

#### Try it yourself

Try to write a combination with the numbers 10, 11, 56, 57, 58.

### 4.2.2. Create a comment with `ctrl + shift + c`

```
# Comments help you organize your code.  
# A line with a comment will not be executed.
```

### 4.2.3. Create a table

A simple table with the number of trips by *car*, *PT*, *walking*, and *cycling* in a hypothetical street segment at a certain period.

#### Define variables

```
modes <- c("car", "PT", "walking", "cycling") # you can use "=" or "<-"
trips = c(200, 50, 300, 150) # key sensitive (uppercase and lowercase
  letters are different)
```

#### Join the variables to create a table

```
table_example = data.frame(modes, trips)
```

#### Take a look at the table

Visualize the table by clicking on the “Data” in the “Environment” page or use :

```
View(table_example)
```

#### Look at the first row

```
table_example[1,] #rows and columns start from 1 in R, differently from
  Python which starts from 0.
```

```
  modes trips
1   car   200
```

#### Look at first row and column

```
table_example[1,1]
```

```
[1] "car"
```

```
table_example[1,4]
```

```
NULL
```

## 4.3. Practical exercise

**Dataset:** the number of trips between all municipalities in the Lisbon Metropolitan Area, Portugal (INE 2018).

### 4.3.1. Import dataset

Download the dataset [TRIPMode\\_mun.Rds](#) and save it in the folder `data`, under your project's root folder.

To import it, you can click directly in the file under the “Files” pan, or:

```
data = readRDS("data/TRIPMode_mun.Rds")
```

💡 After you type " you can use `tab` to navigate between folders and files and `enter` to autocomplete.

### 4.3.2. Take a first look at the data

Summary statistics

```
summary(data)
```

| Origin_mun |          | Destination_mun |         | Total    |           | Walk     |          |
|------------|----------|-----------------|---------|----------|-----------|----------|----------|
| Length     | :315     | Length          | :315    | Min.     | : 7       | Min.     | : 0      |
| N.unique   | : 18     | N.unique        | : 18    | 1st Qu.: | 330       | 1st Qu.: | 0        |
| N.blank    | : 0      | N.blank         | : 0     | Median   | : 1090    | Median   | : 0      |
| Min.nchar: | 5        | Min.nchar:      | 5       | Mean     | : 16825   | Mean     | : 4033   |
| Max.nchar: | 19       | Max.nchar:      | 19      | 3rd Qu.: | 5374      | 3rd Qu.: | 0        |
|            |          |                 |         | Max.     | :875144   | Max.     | :306289  |
| Bike       |          | Car             |         | PTransit |           | Other    |          |
| Min.       | : 0.00   | Min.            | : 0     | Min.     | : 0.0     | Min.     | : 0.0    |
| 1st Qu.:   | 0.00     | 1st Qu.:        | 263     | 1st Qu.: | 5.0       | 1st Qu.: | 0.0      |
| Median     | : 0.00   | Median          | : 913   | Median   | : 134.0   | Median   | : 0.0    |
| Mean       | : 80.19  | Mean            | : 9956  | Mean     | : 2602.6  | Mean     | : 152.4  |
| 3rd Qu.:   | 0.00     | 3rd Qu.:        | 4408    | 3rd Qu.: | 975.5     | 3rd Qu.: | 62.5     |
| Max.       | :5362.00 | Max.            | :349815 | Max.     | :202428.0 | Max.     | :11647.0 |

Check the structure of the data

```
str(data)
```

```
Classes 'tbl_df', 'tbl' and 'data.frame': 315 obs. of 8 variables:
 $ Origin_mun      : chr  "Alcochete" "Alcochete" "Alcochete" "Alcochete" ...
 $ Destination_mun: chr  "Alcochete" "Almada" "Amadora" "Barreiro" ...
 $ Total           : num  20478 567 188 867 114 ...
 $ Walk            : num  6833 0 0 0 0 ...
 $ Bike            : num  320 0 0 0 0 0 0 0 91 0 ...
```

```
$ Car           : num 12484 353 107 861 114 ...
$ PTransit      : num 833 0 81 5 0 ...
$ Other         : num 7 214 0 0 0 0 0 0 0 ...
```

Check the number of rows (observations) and columns (variables)

```
nrow(data)
```

```
[1] 315
```

```
ncol(data)
```

```
[1] 8
```

Open the dataset

```
View(data)
```

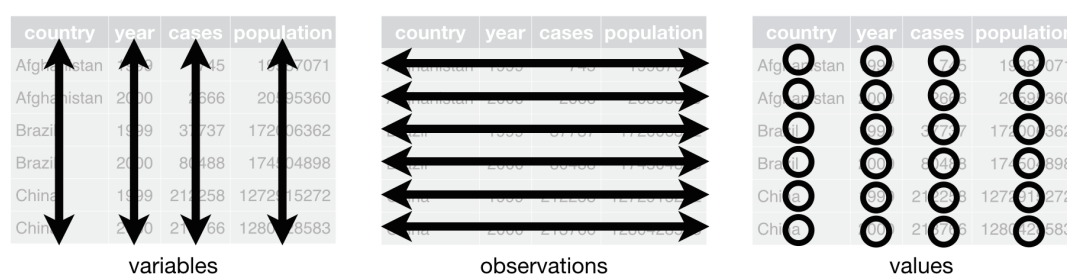


Figure 4.1.: The following three rules make a dataset tidy: variables are columns, observations are rows, and values are cells. [©wickham17r]

### 4.3.3. Explore the data

Check the total number of trips

Use `$` to select a variable of the data

```
sum(data$Total)
```

```
[1] 5299853
```

Percentage of car trips related to the total

#### 4. R basics

```
sum(data$Car)/sum(data$Total) * 100
```

```
[1] 59.17638
```

Percentage of active trips related to the total

```
(sum(data$Walk) + sum(data$Bike)) / sum(data$Total) * 100
```

```
[1] 24.44883
```

#### 4.3.4. Modify original data

Create a column with the sum of the number of trips for active modes

```
data$Active = data$Walk + data$Bike
```

**Filter by condition (create new tables)**

Filter trips only with origin from Lisbon

```
data_Lisbon = data[data$Origin == "Lisboa",]
```

Filter trips with origin **different** from Lisbon

```
data_out_Lisbon = data[data$Origin != "Lisboa",]
```

Filter trips with origin **and** destination in Lisbon

```
data_in_Out_Lisbon = data[data$Origin == "Lisboa" & data$Destination ==  
  "Lisboa",]
```

**Remove the first column**

```
data = data[, -1] #first column
```

**Create a table only with origin, destination and walking trips**

There are many ways to do the same operation.

```
names(data)
```

```
[1] "Destination_mun" "Total"          "Walk"           "Bike"
[5] "Car"             "PTransit"       "Other"          "Active"
```

```
data_walk2 = data[,c(1,2,4)]
```

```
data_walk3 = data[, -c(3,5:9)]
```

### 4.3.5. Export data

Save data in `.csv` and `.Rds`

```
write.csv(data, 'data/dataset.csv', row.names = FALSE)
saveRDS(data, 'data/dataset.Rds') #Choose a different file.
```

### 4.3.6. Import data

```
csv_file = read.csv("data/dataset.csv")
rds_file = readRDS("data/dataset.Rds")
```



## 5. Data manipulation

### 06/10/2026 class materials

Download [class\\_2.zip](#), unzip it and open it on your IDE.

In this chapter we will use some very useful `dplyr` functions to handle and manipulate data.

You can load the `dplyr` package directly, or load the entire tidy universe (`tidyverse`).

```
# library(tidyverse)
library(dplyr)
```

Using the same dataset as in [R basics](#) but with slightly differences<sup>1</sup>.

We will do the same operations but in a simplified way.

Download the dataset [TRIPSTripOrigin.Rds](#) and save it in the folder `data`, under your project's root folder.

```
TRIPS <- readRDS("data/TRIPSTripOrigin.Rds")
```

### Important

Note that it is very important to understand the R basics, that's why we started from there, even if the following functions will provide the same results.

You don't need to know everything! And you don't need to know by heart. The following functions are the ones you will probably use most of the time to handle data.

 There are several ways to reach the same solution. Here we present only one of them.

---

<sup>1</sup>This dataset includes the number of trips with origin in each neighborhood, divided by mode of transport, and inter or intra municipal trips.

## 5.1. Select variables

Have a look at your dataset. You can open using `View()`, look at the information at the “Environment” panel, or even print the same information using `glimpse()`

```
glimpse(TRIPS)
```

We will create a new dataset with *Origin*, *Walk*, *Bike* and *Total*. This time we will use the `select()` function.

```
TRIPS_new <- select(TRIPS, Origin, Walk, Bike, Total) # the first argument  
is the dataset
```

The first argument, as usually in R, is the dataset, and the remaining ones are the columns to select.

With most of the `dplyr` functions you don’t need to refer to `data$...` you can simply type the variable names (and even without the `"..."`!). This makes coding in R simpler :)

You can also remove columns that you don’t need.

```
TRIPS_new <- select(TRIPS_new, -Total) # dropping the Total column
```

### 5.1.1. Using pipes!

Now, let’s introduce pipes. Pipes are a rule as: “**With this, do this.**”

This is useful to skip the first argument of the functions (usually the dataset to apply the function).

Applying a pipe to the `select()` function, we can write as:

```
TRIPS_new <- TRIPS |> select(Origin, Walk, Bike, Total)
```

Two things to **note**:

1. The pipe symbol can be written as `|>` or `%>%`.<sup>2</sup> To write it you may also use the `ctrl+shift+m` shortcut.
2. After typing `select(` you can press `tab` and the list of available variables of that dataset will show up! `Enter` to select. With this you prevent typo errors.

---

<sup>2</sup>You can change this in RStudio > Tools > Global Options > Code.

## 5.2. Filter observations

You can filter observations based on a condition using the `filter()` function.

```
TRIPS2 <- TRIPS[TRIPS$Total > 25000, ] # using r-base, you cant forget the
      comma
TRIPS2 <- TRIPS2 |> filter(Total > 25000) # using dplyr, it's easier
```

You can have other conditions inside the condition.

```
summary(TRIPS$Total)
```

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.   |
|------|---------|--------|-------|---------|--------|
| 361  | 5918    | 17474  | 22457 | 33378   | 112186 |

```
TRIPS3 <- TRIPS |> filter(Total > median(Total))
```

Other filter conditions:

- `==` equal, `!=` different
- `<` smaller, `>` greater, `<=` smaller or equal, `>=` greater or equal
- `&` and, `|` or
- `is.na`, `!is.na` is not NA
- `%in%`, `!%in%` not in

## 5.3. Create new variables

You can also try again to create a variable of Car percentage using pipes! To create a new variable or change an existing one (overwriting), you can use the `mutate()` function.

```
TRIPS$Car_perc <- TRIPS$Car / TRIPS$Total * 100 # using r-base
TRIPS <- TRIPS |> mutate(Car_perc = Car / Total * 100) # using dplyr
```

## 5.4. Change data type

Data can be in different formats. For example, the variable *Origin* is a character, but we can convert it to a numeric variable.

```
class(TRIPS$Origin)
```

## 5. Data manipulation

```
[1] "character"
```

```
TRIPS <- TRIPS |>
  mutate(Origin_num = as.integer(Origin)) # you can use as.numeric() as
  well
class(TRIPS$Origin_num)
```

```
[1] "integer"
```

Most used data types are:

- integer (`int`)
- numeric (`num`)
- character (`chr`)
- logical (`logical`)
- date (`Date`)
- factor (`factor`)

### 5.4.1. Factors

Factors are useful to deal with categorical data. You can convert a character to a factor using `as.factor()`, and also use labels and levels for categorical ordinal data.

We can change the `Lisbon` variable to a factor, and `Internal` too.

```
TRIPS <- TRIPS |>
  mutate(
    Lisbon_factor = factor(Lisbon, labels = c("No", "Yes")),
    Internal_factor = factor(Internal, labels = c("Inter", "Intra"))
  )
```

But how do we know which levels come first? A simple way is to use `table()` or `unique()` functions.

```
unique(TRIPS$Lisbon) # this will show all the different values
```

```
[1] 0 1
```

```
table(TRIPS$Lisbon) # this will show the frequency of each value
```

```
 0  1
188 48
```

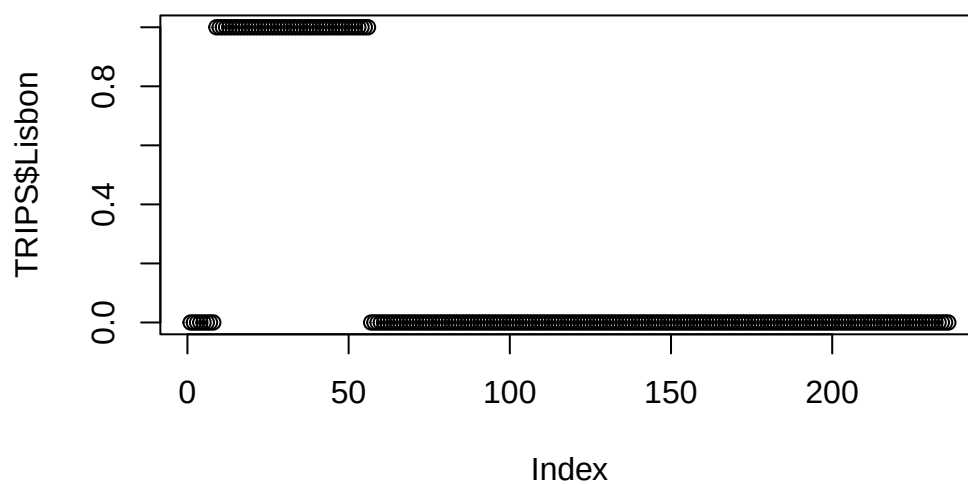
```
table(TRIPS$Lisbon_factor)
```

```
No Yes
188 48
```

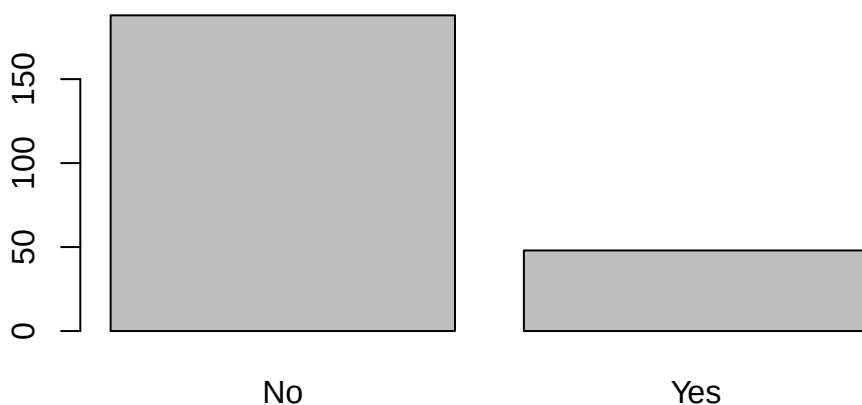
The first number to appear is the first level, and so on.

You can see the difference between using a continuous variable (in this case `Lisbon` has 0 and 1) and a categorical variable (`Lisbon_factor`).

```
plot(TRIPS$Lisbon) # the values range between 0 and 1
```



```
plot(TRIPS$Lisbon_factor) # the values are categorical and labeled with
Yes/No
```



## 5.5. Join data tables

When having relational tables - *i.e.* with a common identifier - it is useful to be able to join them in a very efficient way.

`left_join` is a function that joins two tables by a common column. The first table is the one that will be kept, and the second one will be joined to it. How `left_join` works:

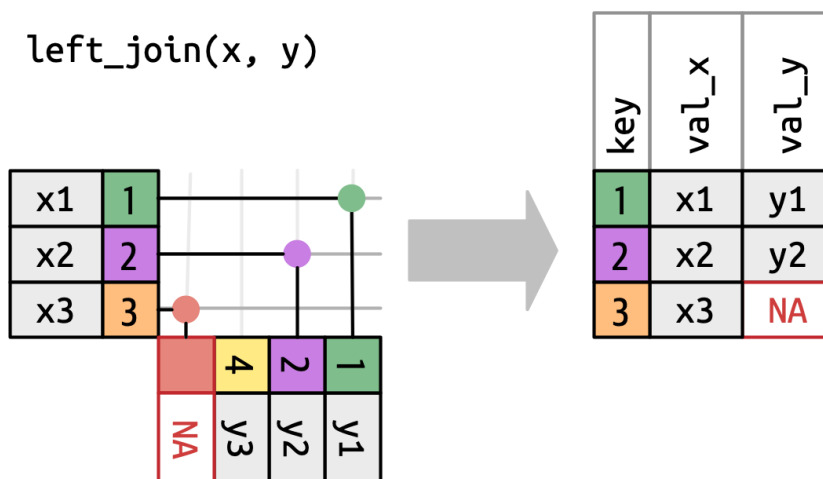


Figure 5.1.: A visual representation of the left join where every row in x appears in the output. Source: R for Data Science.

Let's **join the municipalities** to this table with a supporting table that includes all the **relation** between neighbourhoods and municipalities, and the respective names and codes.

Download the dataset [Municipalities\\_names.Rds](#) and save it in the folder `data`, under your project's root folder.

```
Municipalities <- readRDS("data/Municipalities_names.Rds")
```

```
head(TRIPS)
```

```
# A tibble: 6 x 13
  Origin Total Walk Bike Car PTransit Other Internal Lisbon Car_perc
  <chr>   <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <dbl>
1 110501 35539 11325 1309 21446   1460     0         0         0    60.3
2 110501 47602 3502  416 37727   5519   437         1         0    79.3
3 110506 37183 12645  40 22379   2057    63         0         0    60.2
4 110506 42313 1418  163 37337   3285   106         1         0    88.2
5 110507 30725 9389 1481 19654    201     0         0         0    64.0
6 110507 54586 2630  168 44611   6963   215         1         0    81.7
```

```
# i 3 more variables: Origin_num <int>, Lisbon_factor <fct>,
#   Internal_factor <fct>
```

```
tail(Municipalities)
```

|     | Mun_code | Neighborhood_code | Municipality                                          |
|-----|----------|-------------------|-------------------------------------------------------|
| 113 | 1109     | 110913            | Mafra                                                 |
| 114 | 1114     | 111409            | Vila Franca de Xira                                   |
| 115 | 1109     | 110918            | Mafra                                                 |
| 116 | 1109     | 110904            | Mafra                                                 |
| 117 | 1502     | 150202            | Alcochete                                             |
| 118 | 1109     | 110911            | Mafra                                                 |
|     |          |                   | Neighborhood                                          |
| 113 |          |                   | Santo Isidoro                                         |
| 114 |          |                   | Vila Franca de Xira                                   |
| 115 |          |                   | União das freguesias de Azueira e Sobral da Abelheira |
| 116 |          |                   | Encarnação                                            |
| 117 |          |                   | Samouco                                               |
| 118 |          |                   | Milharado                                             |

We can see that we have a common variable: `Origin` in `TRIPS` and `Neighborhood_code` in `Municipalities`.

To join these two tables we need to specify the common variable in each table, using the `by` argument.

```
TRIPSjoin <- TRIPS |> left_join(Municipalities, by = c("Origin" =
  "Neighborhood_code"))
```

If you prefer, you can mutate or rename a variable so both tables have the same name. When **both tables have the same name**, you don't need to specify the `by` argument.

```
Municipalities <- Municipalities |> rename(Origin = "Neighborhood_code") #
  change name
TRIPSjoin <- TRIPS |> left_join(Municipalities) # automatic detects common
  variable
```

As you can see, both tables don't need to be the same length. The `left_join` function will keep all the observations from the first table, and join the second table to it. If there is no match, the variables from the second table will be filled with `NA`.

## 5.6. group\_by and summarize

We have a very large table with all the neighbourhoods and their respective municipalities. We want to know the total number of trips with origin in each municipality.

To make it easier to understand, let's keep only the variables we need.

```
TRIPSredux <- TRIPSjoin |> select(Origin, Municipality, Internal, Car,
  Total)
head(TRIPSredux)
```

```
# A tibble: 6 x 5
  Origin Municipality Internal    Car Total
  <chr>   <chr>          <dbl> <dbl> <dbl>
1 110501 Cascais             0 21446 35539
2 110501 Cascais             1 37727 47602
3 110506 Cascais             0 22379 37183
4 110506 Cascais             1 37337 42313
5 110507 Cascais             0 19654 30725
6 110507 Cascais             1 44611 54586
```

We can group this table by the `Municipality` variable and summarize the number of trips with origin in each municipality.

```
TRIPSSum <- TRIPSredux |>
  group_by(Municipality) |> # you won't notice any change with only this
  summarize(Total = sum(Total))
head(TRIPSSum)
```

```
# A tibble: 6 x 2
  Municipality Total
  <chr>         <dbl>
1 Alcochete    36789
2 Almada       289834
3 Amadora      344552
4 Barreiro     133658
5 Cascais      373579
6 Lisboa      1365111
```

We summed the total number of trips in each municipality.

If we want to group by more than one variable, we can add more `group_by()` functions.

```
TRIPSSum2 <- TRIPSredux |>
  group_by(Municipality, Internal) |>
  summarize(
    Total = sum(Total),
    Car = sum(Car)
  )
head(TRIPSSum2)
```

```
# A tibble: 6 x 4
# Groups:   Municipality [3]
  Municipality Internal  Total    Car
  <chr>          <dbl> <dbl> <dbl>
1 Alcochete      0  16954  9839
2 Alcochete      1  19835 15632
3 Almada         0 105841 49012
4 Almada         1 183993 125091
5 Amadora        0 117727 33818
6 Amadora        1 226825 142386
```

We summed the total number of trips and car trips in each municipality, **separated by** inter and intra municipal trips.

🔥 It is a good practice to use the `ungroup()` function after the `group_by()` function. This will remove the grouping. If you don't do this, the grouping will be kept and you may have unexpected results in the next time you use that dataset.

## 5.7. Arrange data

You can **sort** a dataset by one or more variables.

For instance, `arrange()` by Total trips, ascending or descending order.

```
TRIPS2 <- TRIPSSum2 |> arrange(Total)
TRIPS2 <- TRIPSSum2 |> arrange(-Total) # descending

TRIPS2 <- TRIPSSum2 |> arrange(Municipality) # alphabetic

TRIPS4 <- TRIPS |> arrange(Lisbon_factor, Total) # more than one variable
```

This is not the same as opening the view table and click on the arrows. When you do that, the order is not saved in the dataset. If you want to save the order, you need to use the `arrange()` function.

## 5.8. All together now!

This is the pipes magic. It takes the last result and applies the next function to it. “With this, do this.”. You can chain as many functions as you want.

```
TRIPS_pipes <- TRIPS |>
  select(Origin, Internal, Car, Total) |>
  mutate(Origin_num = as.integer(Origin)) |>
  mutate(Internal_factor = factor(Internal, labels = c("Inter",
    "Intra"))) |>
  filter(Internal_factor == "Inter") |>
  left_join(Municipalities) |>
  group_by(Municipality) |>
  summarize(
    Total = sum(Total),
    Car = sum(Car),
    Car_perc = Car / Total * 100
  ) |>
  ungroup() |>
  arrange(desc(Car_perc))
```

With this code we will have a table with the total number of intercity trips, by municipality, with their names instead of codes, arranged by the percentage of car trips.

TRIPS\_pipes

```
# A tibble: 18 x 4
  Municipality      Total      Car Car_perc
  <chr>          <dbl>   <dbl>   <dbl>
1 Maфра          65811  46329    70.4
2 Sesimbra       49370  31975    64.8
3 Cascais       161194  96523    59.9
4 Palmela        66428  39688    59.7
5 Alcochete      16954   9839    58.0
6 Setúbal       129059  70318    54.5
7 Montijo        57164  30900    54.1
8 Seixal        120747  63070    52.2
9 Sintra        237445 123408    52.0
10 Oeiras        134862  66972    49.7
11 Almada        105841  49012    46.3
12 Loures        132310  60478    45.7
13 Barreiro       52962  24160    45.6
14 Odivelas      93709  39151    41.8
15 Vila Franca de Xira 115152  47201    41.0
16 Moita          51040  17394    34.1
17 Amadora       117727  33818    28.7
18 Lisboa       280079  69038    24.6
```

## 5.9. Other dplyr functions

You can explore other dplyr functions and variations to manipulate data in the **dplyr** cheat sheet:

### Data transformation with dplyr : : CHEATSHEET



dplyr functions work with pipes and expect tidy data. In tidy data:

Each **variable** is in its own **column** & Each **observation**, or **case**, is in its own **row**  
 $x \gg f(y)$  becomes  $f(x, y)$

#### Summarize Cases

Apply **summary functions** to columns to create a new table of summary statistics. Summary functions take vectors as input and return one value (see back).

**summary function**

```
summarize(data, ...)
mtcars > summarize(avg = mean(mpg))

count(data, ..., wt = NULL, sort = FALSE, name = NULL)
Count number of rows in each group defined by the variables in ... Also tally(), add_count(), add_tally().
mtcars > count(cyl)
```

#### Group Cases

Use **group\_by(data, ...)**, `add = FALSE`, `drop = TRUE` to create a "grouped" copy of a table grouped by columns in ... dplyr functions will manipulate each "group" separately and combine the results.

```
mtcars > group_by(cyl) > summarize(avg = mean(mpg))
```

Use **rowwise(data, ...)** to group data into individual rows. dplyr functions will compute results for each row. Also apply functions to list-columns. See tidy cheat sheet for list-column workflow.

```
starwars > rowwise() > mutate(film_count = length(films))
```

**ungroup(x, ...)** Returns ungrouped copy of table.  
`g_mtcars <- mtcars > group_by(cyl)`  
`ungroup(g_mtcars)`

#### Manipulate Cases

##### EXTRACT CASES

Row functions return a subset of rows as a new table.

```
filter(data, ..., preserve = FALSE) Extract rows that meet logical criteria.
mtcars > filter(mpg > 20)

distinct(data, ..., keep_all = FALSE) Remove rows with duplicate values.
mtcars > distinct(gear)
```

```
slice(data, ..., preserve = FALSE) Select rows by position.
mtcars > slice(10:15)
```

```
slice_sample(data, ..., n, prop, weight_by = NULL, replace = FALSE) Randomly select rows. Use n to select a number of rows and prop to select a fraction of rows.
mtcars > slice_sample(n = 5, replace = TRUE)
```

```
slice_min(data, order_by, ..., n, prop, with_ties = TRUE) and slice_max() Select rows with the lowest and highest values.
mtcars > slice_min(mpg, prop = 0.25)
```

```
slice_head(data, ..., n, prop) and slice_tail() Select the first or last rows.
mtcars > slice_head(n = 5)
```

**Logical and boolean operators to use with filter()**

```
== < <= is.na() %in% | xor()
!= > >= !is.na() ! &
```

See **?base::Logic** and **?Comparison** for help.

##### ARRANGE CASES

```
arrange(data, ..., by, group = FALSE) Order rows by values of a column or columns (low to high), use with desc() to order from high to low.
mtcars > arrange(mpg)
mtcars > arrange(desc(mpg))
```

##### ADD CASES

```
add_row(data, ..., before = NULL, after = NULL) Add one or more rows to a table.
cars > add_row(speed = 1, dist = 1)
```

#### Manipulate Variables

##### EXTRACT VARIABLES

Column functions return a set of columns as a new vector or table.

```
pull(data, var = -1, name = NULL, ...) Extract column values as a vector, by name or index.
mtcars > pull(wt)
```

```
select(data, ...) Extract columns as a table.
mtcars > select(mpg, wt)
```

```
relocate(data, ..., before = NULL, after = NULL) Move columns to new position.
mtcars > relocate(mpg, cyl, after = last_col())
```

Use these helpers with **select()** and **across()**

```
e.g. mtcars > select(mpg:cyl)
contains(match) num_range(prefix, range) ; e.g., mpg:cyl
ends_with(match) all_of(x)/any_of(x, ..., vars) ; e.g., !gear
starts_with(match) matches(match) ; e.g., everything()
```

##### MANIPULATE MULTIPLE VARIABLES AT ONCE

```
df <- tibble(x_1 = c(1, 2), x_2 = c(3, 4), y = c(4, 5))
```

```
across(cols, funs, ..., names = NULL) Summarize or mutate multiple columns in the same way.
df > summarize(across(everything(), mean))
```

```
c_across(cols) Compute across columns in row-wise data.
df > rowwise() > mutate(x_total = sum(c_across(1:2)))
```

##### MAKE NEW VARIABLES

Apply **vectorized functions** to columns. Vectorized functions take vectors as input and return vectors of the same length as output (see back).

```
mutate(data, ..., keep = "all", before = NULL, after = NULL) Compute new column(s). Also add_column().
mtcars > mutate(gpm = 1 / mpg)
mtcars > mutate(gpm = 1 / mpg, keep = "none")
```

```
rename(data, ...) Rename columns. Use rename_with() to rename with a function.
mtcars > rename(miles_per_gallon = mpg)
```



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Take a particular attention to **pivot\_wider** and **pivot\_longer** (**tidyr** package) to transform **OD matrices** in **wide** and **long** formats.

Table 5.1.: OD matrix in long format

| Origins | Destinations | Trips |
|---------|--------------|-------|
| A       | B            | 20    |
| A       | C            | 45    |
| B       | A            | 10    |
| C       | C            | 5     |
| C       | A            | 30    |

Table 5.2.: OD matrix in wide format

| Trips | A  | B  | C  |
|-------|----|----|----|
| A     | NA | 20 | 45 |

## 5. *Data manipulation*

| Trips | A  | B  | C  |
|-------|----|----|----|
| B     | 10 | NA | NA |
| C     | 30 | NA | 5  |

## 6. Exploratory Data Analysis

### 6.1. Intro

This chapter will show you how to use visualization and transformation to explore your data in a systematic way, a task that statisticians call exploratory data analysis, or **EDA** for short. EDA is an iterative cycle, where you should feel free to “fall in love” with your data.

You:

1. Generate questions about your data.
2. Search for answers by visualizing, transforming, and modelling your data.
3. Use what you learn to refine your questions and/or generate new questions.

Note: EDA is not a formal process with a strict set of rules!

See Wickham et al. (2017) for more, in particular the EDA chapter.

#### Do it yourself with R

Copy the script [ExploratoryDataAnalysis.R](#) and paste it in your session.  
Run each line using CTRL + ENTER

### 6.2. Load packages

```
library(tidyverse) # Pack of most used libraries for data science
library(skimr) # summary of the data
library(DataExplorer) # exploratory data analysis
library(corrplot) # correlation plots
```

### 6.3. Dataset

The database used in this example is a treated database from the Mobility Survey for the metropolitan areas of Lisbon and Porto in 2018 (INE 2018). We will only focus on trips within the metropolitan area of Lisbon.

Included **variables**:

## 6. Exploratory Data Analysis

- `Origin_dicofre16` - Code of Freguesia (district) as set by INE after 2016 (Distrito + Concelho + Freguesia), for trip origin
- `Total` - number of trips with origin at each district
- `Walk` - number of walking trips
- `Bike` - number of bike trips
- `Car` - number of car trips. Includes taxi and motorcycle.
- `PTransit` - number of Public Transit trips
- `Other` - number of other trips (truck, van, tractor, aviation)
- `Distance` - average trip distance (km)
- `Duration` - average trip duration (minutes)
- `Car_perc` - percentage of car trips
- `N_INDIVIDUOS` - number of residents (INE 2022)
- `Male_perc` - percentage of male residents (INE 2022)
- `IncomeHH` - average household income
- `Nvehicles` - average number of car/motorcycle vehicles in the household
- `DrivingLic` - percentage of car driving licence holders
- `CarParkFree_Work` - percentage of respondents with free car parking at the work location
- `PTpass` - percentage of public transit monthly pass holders
- `internal` - binary variable (factor). **Yes**: trip with same TAZ origin and destination, **No**: trips with different destination
- `Lisboa` - binary variable (factor). **Yes**: the district is part of Lisbon municipality, **No**: otherwise
- `Area_km2` - area of in `Origin_dicofre16`, in km<sup>2</sup>

### 6.3.1. Import dataset

Download the dataset [IMOBmodel.Rds](#) and save it in the folder `data`, under your project's root folder.

```
data = readRDS("data/IMOBmodel.Rds")
```

Take a look at the dataset

```
View(data) # open in table
glimpse(data) # glimpse of the dataset
```

```
Rows: 236
Columns: 20
$ Origin_dicofre16 <chr> "110501", "110501", "110506", "110506", "110507", "11~
$ Total           <dbl> 35539, 47602, 37183, 42313, 30725, 54586, 57747, 6788~
$ Walk           <dbl> 11325, 3502, 12645, 1418, 9389, 2630, 20423, 3573, 62~
$ Bike           <dbl> 1309, 416, 40, 163, 1481, 168, 1406, 809, 13, 37, 3, ~
$ Car            <dbl> 21446, 37727, 22379, 37337, 19654, 44611, 33044, 5225~
$ PTransit       <dbl> 1460, 5519, 2057, 3285, 201, 6963, 2477, 10534, 110, ~
```

```

$ Other          <dbl> 0, 437, 63, 106, 0, 215, 396, 717, 2, 169, 3, 824, 0, ~
$ Distance       <dbl> 11.779, 11.779, 9.868, 9.868, 9.600, 9.600, 12.875, 1~
$ Duration       <dbl> 23.96, 23.96, 22.78, 22.78, 23.39, 23.39, 26.89, 26.8~
$ Car_perc       <dbl> 60.34497, 79.25507, 60.18611, 88.24002, 63.96745, 81.~
$ N_INDIVIDUOS   <dbl> 44165, 44165, 59238, 59238, 46529, 46529, 64192, 6419~
$ Male_perc      <dbl> 47.66, 47.66, 47.12, 47.12, 45.54, 45.54, 46.14, 46.1~
$ IncomeHH       <dbl> 2204.32, 2204.32, 1896.05, 1896.05, 2430.43, 2430.43, ~
$ Nvehicles      <dbl> 1.86, 1.86, 1.73, 1.73, 1.60, 1.60, 1.64, 1.64, 1.20, ~
$ DrivingLic     <dbl> 63.03, 63.03, 65.42, 65.42, 67.05, 67.05, 72.75, 72.7~
$ CarParkFree_Work <dbl> 49.97, 49.97, 57.09, 57.09, 47.43, 47.43, 56.47, 56.4~
$ PTpass        <dbl> 11.14, 11.14, 17.12, 17.12, 18.25, 18.25, 19.73, 19.7~
$ internal       <fct> Yes, No, Yes, No, Yes, No, Yes, No, Yes, No, Yes, No, ~
$ Lisboa        <fct> No, No, No, No, No, No, No, No, Yes, Yes, Yes, Yes, Y~
$ Area_km2       <dbl> 39.767414, 39.767414, 20.364310, 20.364310, 8.109861, ~

```

```
str(data) # Structure of the dataset
```

```

'data.frame':  236 obs. of  20 variables:
 $ Origin_dicofre16: chr  "110501" "110501" "110506" "110506" ...
 $ Total           : num  35539 47602 37183 42313 30725 ...
 $ Walk           : num  11325 3502 12645 1418 9389 ...
 $ Bike           : num  1309 416 40 163 1481 ...
 $ Car            : num  21446 37727 22379 37337 19654 ...
 $ PTransit       : num  1460 5519 2057 3285 201 ...
 $ Other          : num  0 437 63 106 0 215 396 717 2 169 ...
 $ Distance       : num  11.78 11.78 9.87 9.87 9.6 ...
 $ Duration       : num  24 24 22.8 22.8 23.4 ...
 $ Car_perc       : num  60.3 79.3 60.2 88.2 64 ...
 $ N_INDIVIDUOS   : num  44165 44165 59238 59238 46529 ...
 $ Male_perc      : num  47.7 47.7 47.1 47.1 45.5 ...
 $ IncomeHH       : num  2204 2204 1896 1896 2430 ...
 $ Nvehicles      : num  1.86 1.86 1.73 1.73 1.6 1.6 1.64 1.64 1.2 1.2 ...
 $ DrivingLic     : num  63 63 65.4 65.4 67 ...
 $ CarParkFree_Work: num  50 50 57.1 57.1 47.4 ...
 $ PTpass        : num  11.1 11.1 17.1 17.1 18.2 ...
 $ internal       : Factor w/ 2 levels "Yes","No": 1 2 1 2 1 2 1 2 1 2 ...
 $ Lisboa        : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 2 2 ...
 $ Area_km2       : num  39.77 39.77 20.36 20.36 8.11 ...

```

## 6.4. Summary statistics

Have an overview of the variables and their statistics' summary.

```
summary(data) # Check the summary statistics
```

## 6. Exploratory Data Analysis

```

Origin_dicofre16      Total      Walk      Bike
Length :236      Min. : 361      Min. : 0.0      Min. : 0.00
N.unique :118      1st Qu.: 5918      1st Qu.: 763.2      1st Qu.: 0.00
N.blank : 0      Median : 17474      Median : 3125.0      Median : 13.50
Min.nchar: 6      Mean : 22457      Mean : 5383.4      Mean : 107.03
Max.nchar: 6      3rd Qu.: 33378      3rd Qu.: 8298.5      3rd Qu.: 98.75
Max. :112186      Max. :32646.0      Max. :2040.00

      Car      PTransit      Other      Distance
Min. : 0      Min. : 0      Min. : 0.0      Min. : 6.835
1st Qu.: 3243      1st Qu.: 249      1st Qu.: 2.0      1st Qu.: 9.539
Median : 9008      Median : 1057      Median : 44.0      Median :10.323
Mean :13289      Mean : 3474      Mean : 203.4      Mean :11.139
3rd Qu.:21249      3rd Qu.: 4853      3rd Qu.: 281.5      3rd Qu.:12.097
Max. :52631      Max. :41672      Max. :2391.0      Max. :22.660

Duration      Car_perc      N_INDIVIDUOS      Male_perc
Min. :16.30      Min. : 0.00      Min. : 1566      Min. :44.61
1st Qu.:23.00      1st Qu.:45.40      1st Qu.:11060      1st Qu.:46.58
Median :24.70      Median :62.62      Median :20855      Median :47.51
Mean :25.42      Mean :59.00      Mean :24324      Mean :47.54
3rd Qu.:27.73      3rd Qu.:75.72      3rd Qu.:36079      3rd Qu.:48.29
Max. :37.42      Max. :99.27      Max. :68649      Max. :55.94

IncomeHH      Nvehicles      DrivingLic      CarParkFree_Work
Min. : 884.5      Min. :1.020      Min. :37.04      Min. : 5.47
1st Qu.:1417.8      1st Qu.:1.350      1st Qu.:57.67      1st Qu.:40.39
Median :1594.7      Median :1.545      Median :63.00      Median :50.51
Mean :1732.6      Mean :1.530      Mean :62.50      Mean :49.30
3rd Qu.:1953.5      3rd Qu.:1.670      3rd Qu.:68.84      3rd Qu.:57.92
Max. :3462.3      Max. :2.420      Max. :80.79      Max. :87.60

PTpass      internal      Lisboa      Area_km2
Min. : 0.00      Yes:118      No :188      Min. : 1.494
1st Qu.:13.41      No :118      Yes: 48      1st Qu.: 5.044
Median :22.71      Mean : 11.598
Mean :23.82      Mean : 25.553
3rd Qu.:32.94      3rd Qu.: 28.511
Max. :60.45      Max. :282.125

```

```
skim(data) # In a more organized way
```

Table 6.1.: Data summary

|                        |      |
|------------------------|------|
| Name                   | data |
| Number of rows         | 236  |
| Number of columns      | 20   |
| Column type frequency: |      |
| character              | 1    |
| factor                 | 2    |
| numeric                | 17   |
| Group variables        | None |

Variable type: character

| skim_variable    | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|------------------|-----------|---------------|-----|-----|-------|----------|------------|
| Origin_dicofre16 | 0         | 1             | 6   | 6   | 0     | 118      | 0          |

Variable type: factor

| skim_variable | n_missing | complete_rate | ordered | n_unique | top_counts        |
|---------------|-----------|---------------|---------|----------|-------------------|
| internal      | 0         | 1             | FALSE   | 2        | Yes: 118, No: 118 |
| Lisboa        | 0         | 1             | FALSE   | 2        | No: 188, Yes: 48  |

Variable type: numeric

| skim_variable | n_miss-<br>ing | com-<br>plete_rate | mean     | sd       | p0     | p25     | p50      | p75      | p100      | hist |
|---------------|----------------|--------------------|----------|----------|--------|---------|----------|----------|-----------|------|
| Total         | 0              | 1                  | 22457.00 | 19084.45 | 361.00 | 5917.75 | 17474.00 | 33377.50 | 112186.00 |      |
| Walk          | 0              | 1                  | 5383.44  | 6224.84  | 0.00   | 763.25  | 3125.00  | 8298.50  | 32646.00  |      |
| Bike          | 0              | 1                  | 107.03   | 248.65   | 0.00   | 0.00    | 13.50    | 98.75    | 2040.00   |      |
| Car           | 0              | 1                  | 13289.24 | 12351.61 | 0.00   | 3243.00 | 9008.00  | 21248.75 | 52631.00  |      |
| PTransit      | 0              | 1                  | 3473.79  | 5467.82  | 0.00   | 249.00  | 1057.00  | 4853.00  | 41672.00  |      |
| Other         | 0              | 1                  | 203.45   | 336.04   | 0.00   | 2.00    | 44.00    | 281.50   | 2391.00   |      |
| Distance      | 0              | 1                  | 11.14    | 2.66     | 6.84   | 9.54    | 10.32    | 12.10    | 22.66     |      |
| Duration      | 0              | 1                  | 25.42    | 3.91     | 16.30  | 23.00   | 24.70    | 27.73    | 37.42     |      |

| skim_variable | n_missing | complete_rate | mean     | sd       | p0      | p25      | p50      | p75      | p100     | hist |
|---------------|-----------|---------------|----------|----------|---------|----------|----------|----------|----------|------|
| Car_perc      | 0         | 1             | 59.00    | 21.56    | 0.00    | 45.40    | 62.62    | 75.72    | 99.27    |      |
| N_INDIVID-UOS | 0         | 1             | 24323.80 | 16438.04 | 1566.00 | 11060.00 | 20855.00 | 36079.00 | 68649.00 |      |
| Male_perc     | 0         | 1             | 47.54    | 1.57     | 44.61   | 46.58    | 47.50    | 48.29    | 55.94    |      |
| IncomeHH      | 0         | 1             | 1732.55  | 453.11   | 884.46  | 1417.76  | 1594.73  | 1953.50  | 3462.32  |      |
| Nvehicles     | 0         | 1             | 1.53     | 0.24     | 1.02    | 1.35     | 1.54     | 1.67     | 2.42     |      |
| DrivingLic    | 0         | 1             | 62.50    | 8.12     | 37.04   | 57.67    | 63.00    | 68.84    | 80.79    |      |
| CarPark-      | 0         | 1             | 49.30    | 14.30    | 5.47    | 40.39    | 50.51    | 57.92    | 87.60    |      |
| Free_Work     |           |               |          |          |         |          |          |          |          |      |
| PTpass        | 0         | 1             | 23.82    | 12.87    | 0.00    | 13.41    | 22.72    | 32.94    | 60.45    |      |
| Area_km2      | 0         | 1             | 25.55    | 42.58    | 1.49    | 5.04     | 11.60    | 28.51    | 282.13   |      |

## 6.5. Missing data

Is there missing data (NA)? How many?

```
table(is.na(data))
```

FALSE

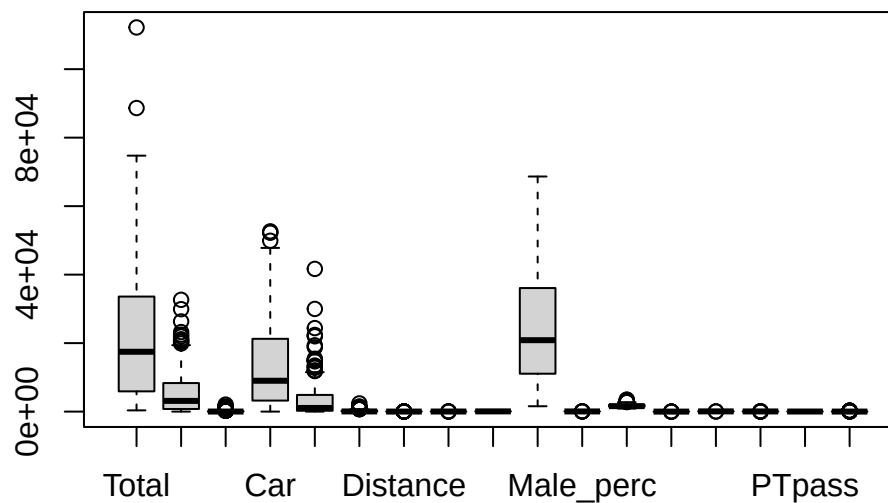
4720

## 6.6. Outliers

Inspect outliers and distributions

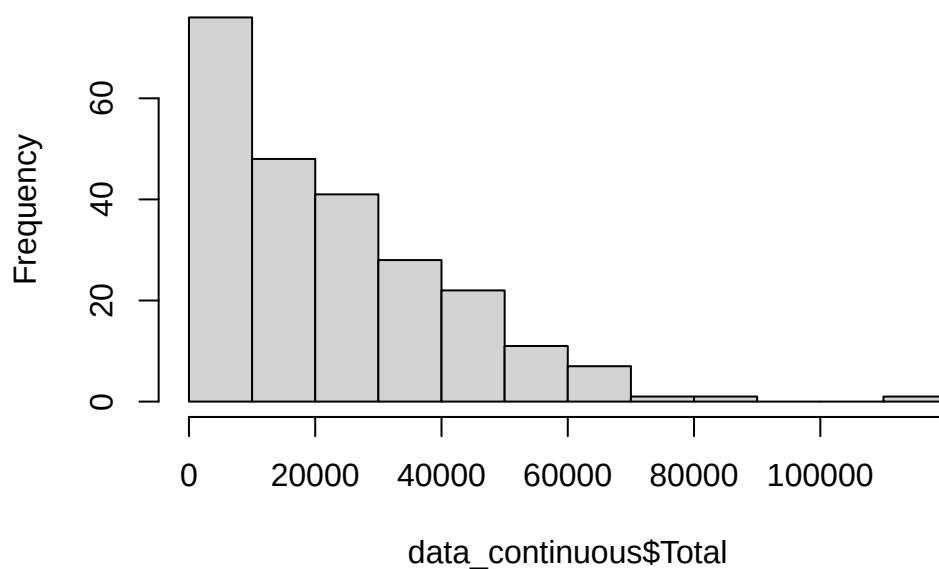
```
boxplot(data) # This does not work if variables are not all continuous
```

```
data_continuous = data |> select(-Origin_dicofre16, -internal, -Lisboa) #  
  Exclude categorical variables  
boxplot(data_continuous) # Exclude categorical variables
```

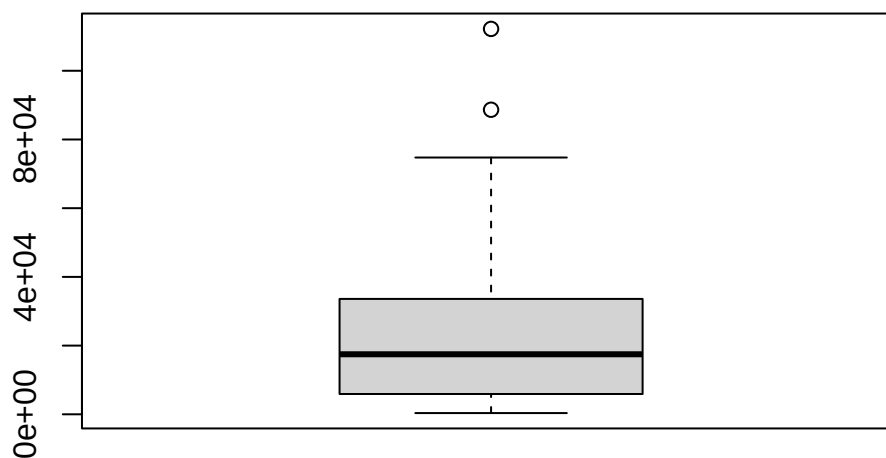


```
hist(data_continuous$Total) # histogram
```

**Histogram of data\_continuous\$Total**



```
boxplot(data_continuous$Total) # outliers detected
```



### 6.6.1. Treating outliers

Based on the theory, we can create a function to detect the outliers.

```
outlier = function(x) {  
  q = quantile(x, probs = c(0.25, 0.75), na.rm = TRUE) # Q1 and Q3  
  caps = quantile(x, probs = c(0.05, 0.95), na.rm = TRUE) # 5th and 95th  
  percentile  
}
```

```
H = 1.5 * IQR(x, na.rm = TRUE) # interquartile range

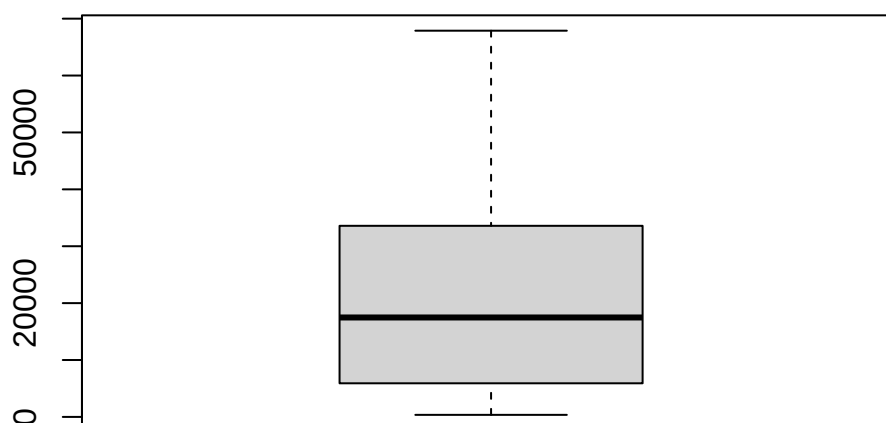
case_when(
  x < (q[1] - H) ~ caps[1], # replace values that are LESS than
  Q1-1.5*IQR with the P5 value
  x > (q[2] + H) ~ caps[2], # replace values that are MORE than
  Q3+1.5*IQR with the P95 value
  TRUE ~ x # otherwise, return the original value
)
}
```

Now, use it in a copy of the table

```
data_outliers = data_continuous # duplicate the table
data_outliers$Total = outlier(data_outliers$Total) # Use the function to
the same variable
```

Take a look again at the boxplot

```
boxplot(data_outliers$Total)
```



Compare the variable with and without the outliers

```
# Mean
mean(data$Total)
```

```
[1] 22457
```

```
mean(data_outliers$Total)
```

## 6. Exploratory Data Analysis

```
[1] 22013.51
```

```
# Median  
median(data$Total)
```

```
[1] 17474
```

```
median(data_outliers$Total)
```

```
[1] 17474
```

```
# Standard deviation  
sd(data$Total)
```

```
[1] 19084.45
```

```
sd(data_outliers$Total)
```

```
[1] 17739.39
```

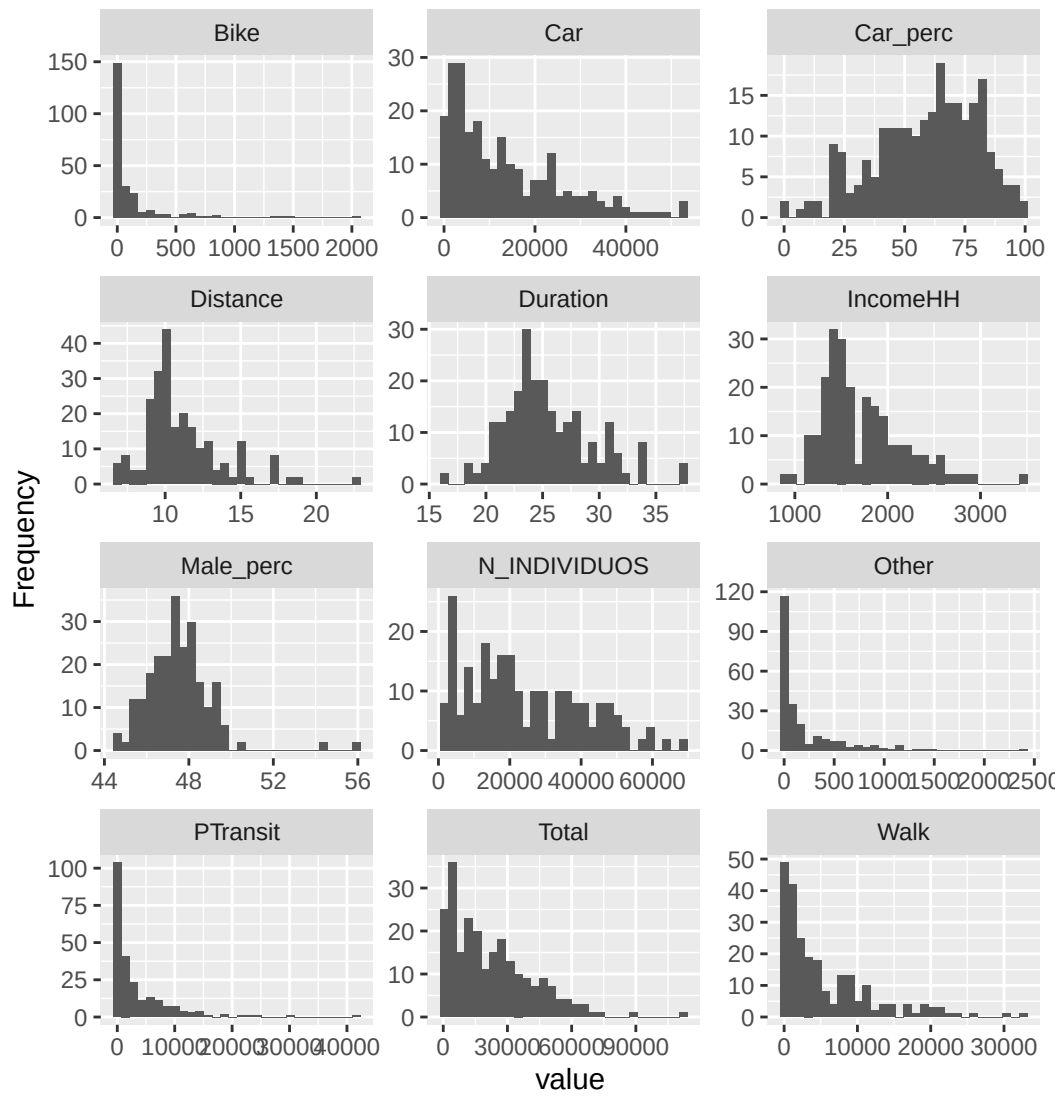
### **i** Note

There are many methods to treat outliers. This is just one of them.

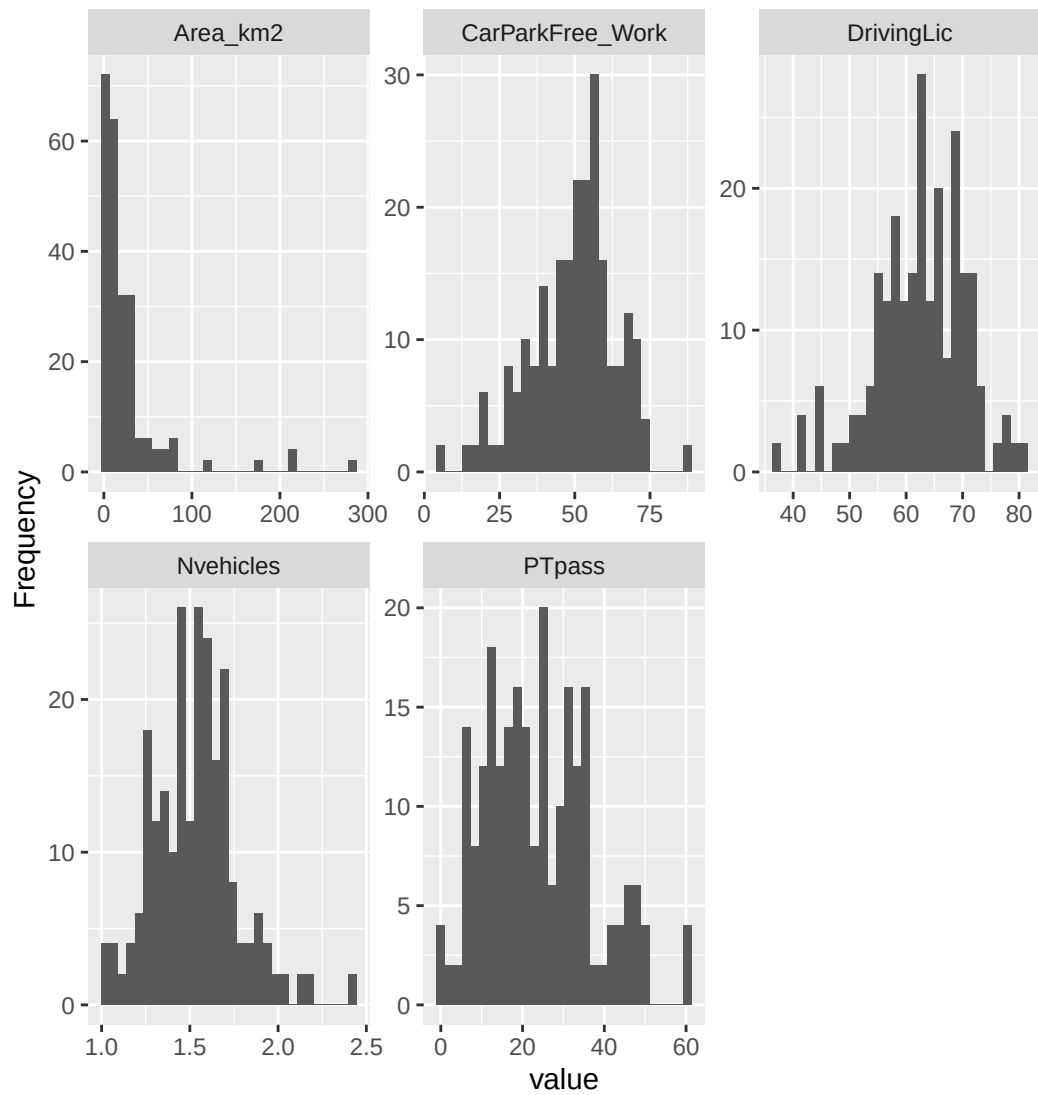
## 6.7. Histograms

Plot histograms of all the continuous variables

```
plot_histogram(data, ncol = 3) # histograms with 3 columns
```

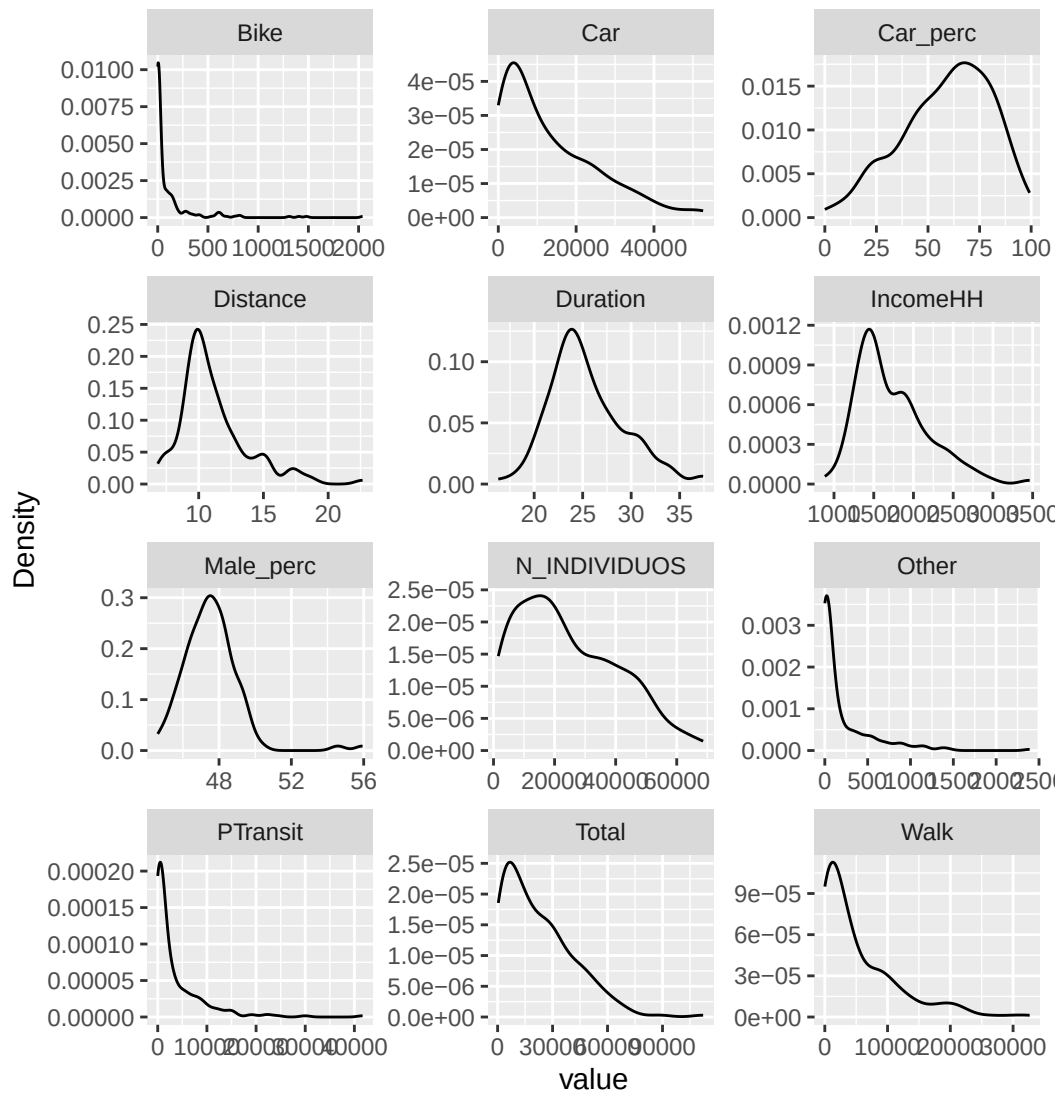


## 6. Exploratory Data Analysis

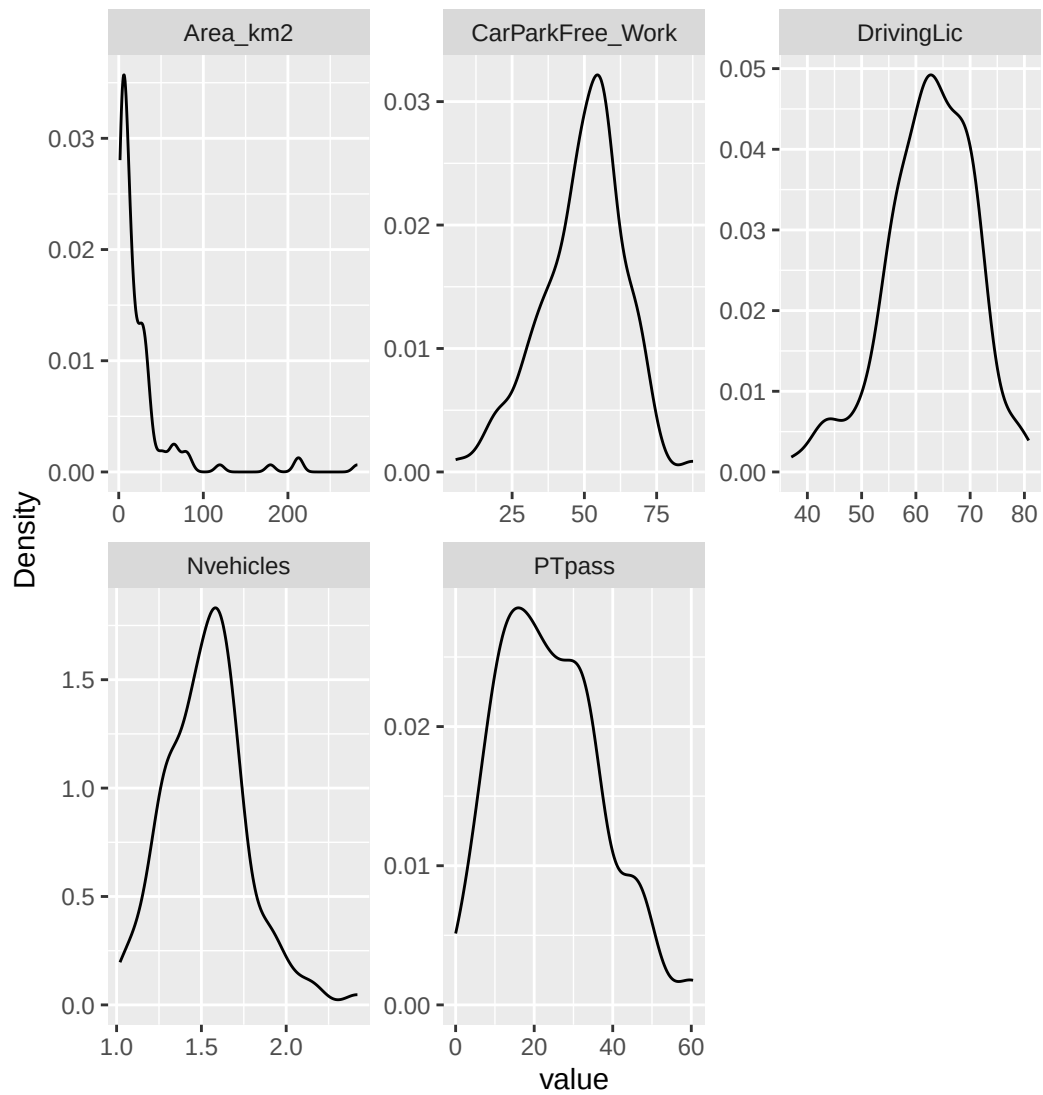


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```
plot_density(data, ncol = 3) # density plots
```



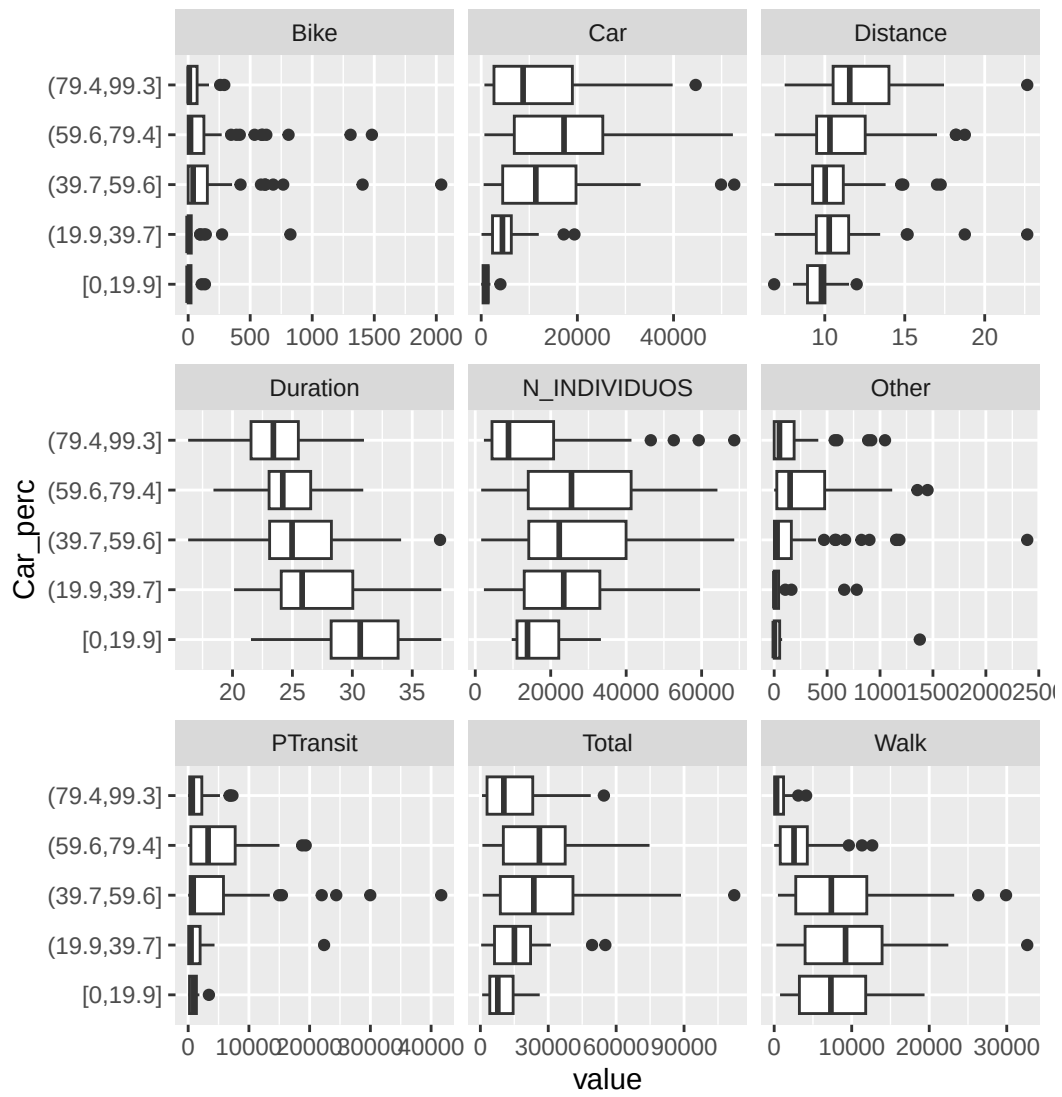
## 6. Exploratory Data Analysis



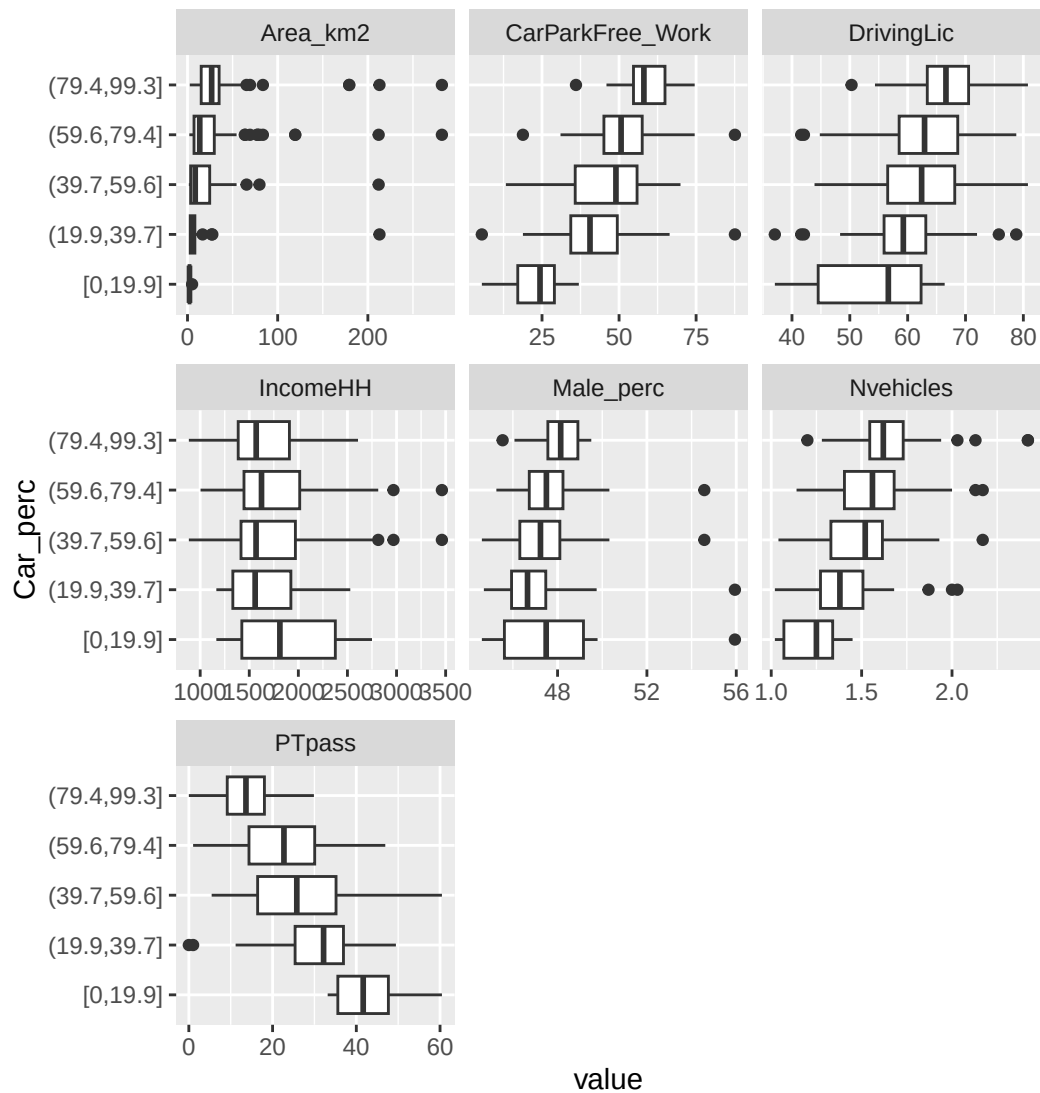
Page 2

Check how other variables are distributed regarding `Car_perc`

```
plot_boxplot(data, by = "Car_perc", ncol = 3)
```



## 6. Exploratory Data Analysis



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### **i** Note

When Car\_perc increases, PTpass decreases.

### **💡** Exercise

Try plotting the boxplots of each independent variable with Car\_perc.

## 6.8. Correlations

Plot correlation heatmaps, between continuous variables.

## 6. Exploratory Data Analysis

```
# estimate correlation matrix
corrmat = cor(data_continuous, method = "pearson") |> round(2)
corrmat
```

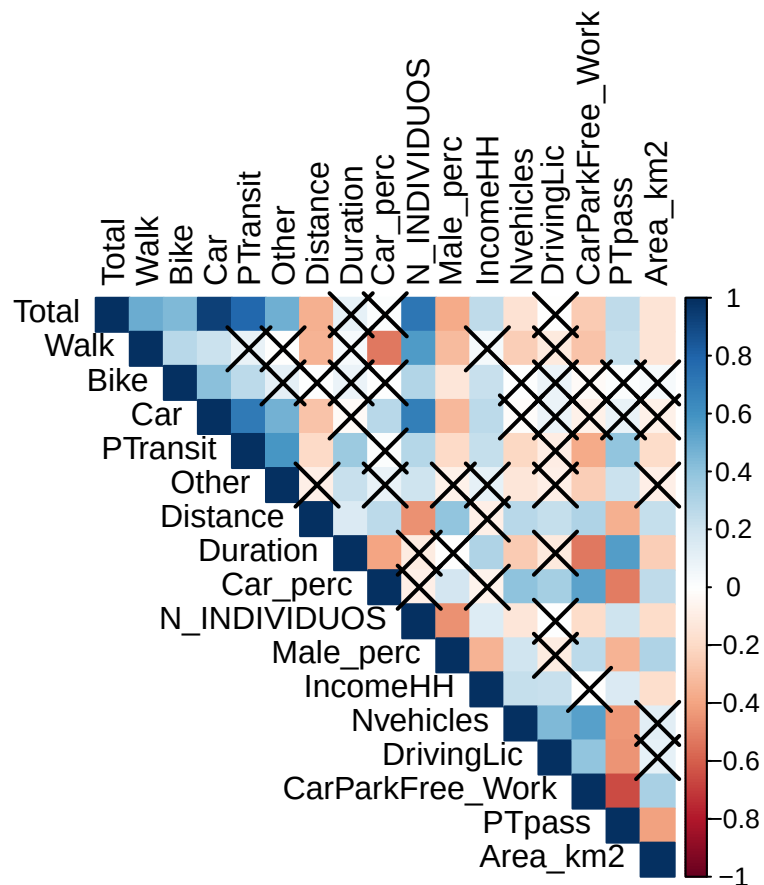
|                  | Total    | Walk         | Bike      | Car      | PTransit  | Other      | Distance | Duration |
|------------------|----------|--------------|-----------|----------|-----------|------------|----------|----------|
| Total            | 1.00     | 0.49         | 0.44      | 0.93     | 0.78      | 0.48       | -0.36    | 0.10     |
| Walk             | 0.49     | 1.00         | 0.27      | 0.21     | 0.09      | -0.01      | -0.34    | 0.03     |
| Bike             | 0.44     | 0.27         | 1.00      | 0.41     | 0.26      | 0.11       | 0.00     | 0.08     |
| Car              | 0.93     | 0.21         | 0.41      | 1.00     | 0.70      | 0.47       | -0.29    | -0.03    |
| PTransit         | 0.78     | 0.09         | 0.26      | 0.70     | 1.00      | 0.58       | -0.20    | 0.36     |
| Other            | 0.48     | -0.01        | 0.11      | 0.47     | 0.58      | 1.00       | -0.09    | 0.22     |
| Distance         | -0.36    | -0.34        | 0.00      | -0.29    | -0.20     | -0.09      | 1.00     | 0.16     |
| Duration         | 0.10     | 0.03         | 0.08      | -0.03    | 0.36      | 0.22       | 0.16     | 1.00     |
| Car_perc         | 0.01     | -0.53        | 0.00      | 0.27     | 0.01      | 0.09       | 0.26     | -0.40    |
| N_INDIVIDUOS     | 0.72     | 0.56         | 0.29      | 0.68     | 0.28      | 0.20       | -0.46    | -0.11    |
| Male_perc        | -0.38    | -0.32        | -0.14     | -0.33    | -0.20     | -0.08      | 0.39     | -0.01    |
| IncomeHH         | 0.25     | 0.04         | 0.22      | 0.26     | 0.23      | 0.10       | -0.09    | 0.30     |
| Nvehicles        | -0.16    | -0.25        | 0.00      | -0.02    | -0.21     | -0.14      | 0.27     | -0.26    |
| DrivingLic       | -0.01    | -0.12        | 0.08      | 0.08     | -0.10     | -0.09      | 0.23     | -0.12    |
| CarParkFree_Work | -0.26    | -0.29        | -0.03     | -0.08    | -0.38     | -0.25      | 0.30     | -0.53    |
| PTpass           | 0.25     | 0.23         | 0.01      | 0.09     | 0.39      | 0.21       | -0.36    | 0.55     |
| Area_km2         | -0.15    | -0.15        | 0.03      | -0.08    | -0.19     | -0.09      | 0.23     | -0.25    |
|                  | Car_perc | N_INDIVIDUOS | Male_perc | IncomeHH | Nvehicles | DrivingLic |          |          |
| Total            | 0.01     | 0.72         | -0.38     | 0.25     | -0.16     | -          |          |          |
| 0.01             |          |              |           |          |           |            |          |          |
| Walk             | -0.53    | 0.56         | -0.32     | 0.04     | -0.25     | -          |          |          |
| 0.12             |          |              |           |          |           |            |          |          |
| Bike             | 0.00     | 0.29         | -0.14     | 0.22     | 0.00      | 0.08       |          |          |
| Car              | 0.27     | 0.68         | -0.33     | 0.26     | -0.02     | 0.08       |          |          |
| PTransit         | 0.01     | 0.28         | -0.20     | 0.23     | -0.21     | -          |          |          |
| 0.10             |          |              |           |          |           |            |          |          |
| Other            | 0.09     | 0.20         | -0.08     | 0.10     | -0.14     | -          |          |          |
| 0.09             |          |              |           |          |           |            |          |          |
| Distance         | 0.26     | -0.46        | 0.39      | -0.09    | 0.27      | 0.23       |          |          |
| Duration         | -0.40    | -0.11        | -0.01     | 0.30     | -0.26     | -          |          |          |
| 0.12             |          |              |           |          |           |            |          |          |
| Car_perc         | 1.00     | -0.09        | 0.18      | -0.07    | 0.40      | 0.33       |          |          |
| N_INDIVIDUOS     | -0.09    | 1.00         | -0.46     | 0.14     | -0.14     | 0.00       |          |          |
| Male_perc        | 0.18     | -0.46        | 1.00      | -0.34    | 0.19      | -          |          |          |
| 0.11             |          |              |           |          |           |            |          |          |
| IncomeHH         | -0.07    | 0.14         | -0.34     | 1.00     | 0.23      | 0.22       |          |          |
| Nvehicles        | 0.40     | -0.14        | 0.19      | 0.23     | 1.00      | 0.44       |          |          |
| DrivingLic       | 0.33     | 0.00         | -0.11     | 0.22     | 0.44      | 1.00       |          |          |
| CarParkFree_Work | 0.53     | -0.19        | 0.26      | -0.01    | 0.54      | 0.39       |          |          |
| PTpass           | -0.52    | 0.20         | -0.34     | 0.16     | -0.44     | -          |          |          |
| 0.45             |          |              |           |          |           |            |          |          |

## 6. Exploratory Data Analysis

|                  |                  |        |          |       |      |      |
|------------------|------------------|--------|----------|-------|------|------|
| Area_km2         | 0.25             | -0.19  | 0.30     | -0.18 | 0.12 | 0.10 |
|                  | CarParkFree_Work | PTpass | Area_km2 |       |      |      |
| Total            | -0.26            | 0.25   | -0.15    |       |      |      |
| Walk             | -0.29            | 0.23   | -0.15    |       |      |      |
| Bike             | -0.03            | 0.01   | 0.03     |       |      |      |
| Car              | -0.08            | 0.09   | -0.08    |       |      |      |
| PTransit         | -0.38            | 0.39   | -0.19    |       |      |      |
| Other            | -0.25            | 0.21   | -0.09    |       |      |      |
| Distance         | 0.30             | -0.36  | 0.23     |       |      |      |
| Duration         | -0.53            | 0.55   | -0.25    |       |      |      |
| Car_perc         | 0.53             | -0.52  | 0.25     |       |      |      |
| N_INDIVIDUOS     | -0.19            | 0.20   | -0.19    |       |      |      |
| Male_perc        | 0.26             | -0.34  | 0.30     |       |      |      |
| IncomeHH         | -0.01            | 0.16   | -0.18    |       |      |      |
| Nvehicles        | 0.54             | -0.44  | 0.12     |       |      |      |
| DrivingLic       | 0.39             | -0.45  | 0.10     |       |      |      |
| CarParkFree_Work | 1.00             | -0.66  | 0.32     |       |      |      |
| PTpass           | -0.66            | 1.00   | -0.41    |       |      |      |
| Area_km2         | 0.32             | -0.41  | 1.00     |       |      |      |

```
# store the results so you can call the p-value at the corrplot
res = cor.mtest(data_continuous, conf.level = .95)

corrplot(
  corrmat,
  method = "color", # or "circle"
  p.mat = res$p,
  sig.level = 0.05,
  type = "upper", # display only the upper triangular
  # order = "hclust", # order by hierarchical clustering
  tl.col = "black" # text label color
)
```



**i Note**

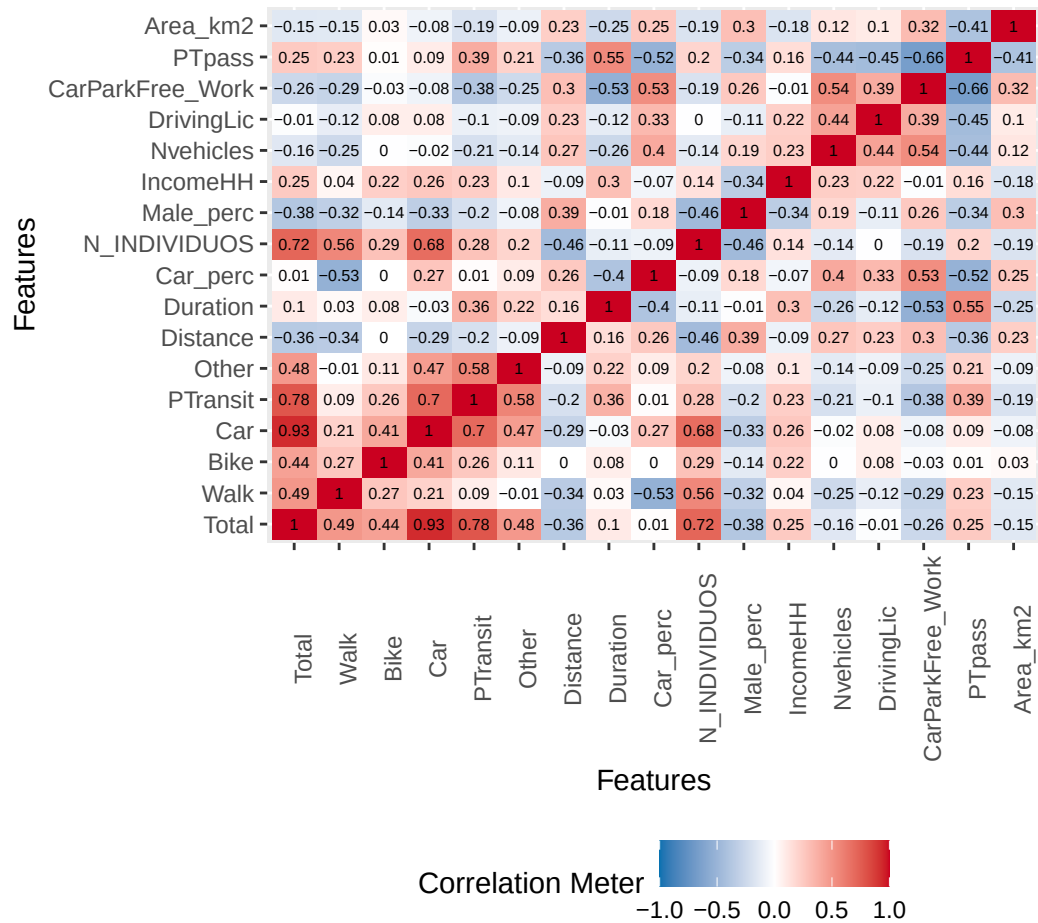
The pairwise correlations that are crossed are statistically not significant. The null hypothesis ( $H_0$ ) is that correlation is zero. The alternative hypothesis ( $H_A$ ) is that correlation is not zero.

This means that the correlations are only significant when you reject the null hypothesis ( $p\text{-value} < 0.05$ ).

Other method, using the `DataExplorer` package

```
plot_correlation(data_continuous)
```

## 6. Exploratory Data Analysis



Check the *p-value* of a crossed pair correlation:

```
cor.test(data$IncomeHH, data$Bike)
```

Pearson's product-moment correlation

```
data: data$IncomeHH and data$Bike
t = 3.5261, df = 234, p-value = 0.0005074
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.09977779 0.34249284
sample estimates:
cor
0.2246163
```

```
cor.test(data$IncomeHH, data$Duration)
```

Pearson's product-moment correlation

```
data: data$IncomeHH and data$Duration
t = 4.817, df = 234, p-value = 2.62e-06
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1795420 0.4122446
sample estimates:
      cor
0.300356
```

```
cor.test(data$Distance, data$Duration)
```

Pearson's product-moment correlation

```
data: data$Distance and data$Duration
t = 2.4605, df = 234, p-value = 0.0146
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.0317533 0.2808153
sample estimates:
      cor
0.1588098
```

💡 The default for `cor.test` is Pearson, two-sided, with a 95% confident level. Check `?cor.test` for more options.



## References

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